

The Binomial Theorem

ANSWER KEY



Binomial coefficients

- 1 Evaluate each binomial coefficient:

a $\binom{6}{2} = 15$

b $\binom{8}{0} = 1$

c $\binom{5}{5} = 1$

d $\binom{10}{3} = 120$

- 2 Write down row 4 of Pascal's triangle (the row beginning 1, 4, ...) and explain how it relates to the expansion of $(a + b)^4$.

Row 4 is 1, 4, 6, 4, 1 — these are the coefficients of $a^4, a^3b, a^2b^2, ab^3, b^4$ in the expansion.

Expansions and specific terms

- 3 Expand $(x + 2)^4$ fully.

$$x^4 + 8x^3 + 24x^2 + 32x + 16.$$

- 4 Find the coefficient of x^3 in the expansion of $(2x - 1)^5$.

Term: $\binom{5}{2}(2x)^3(-1)^2 = 10 \cdot 8x^3 = 80x^3$. Coefficient: 80.

- 5 Find the term containing y^4 in the expansion of $(x + y)^7$.

$$\binom{7}{4}x^3y^4 = 35x^3y^4.$$

- 6 Use the binomial theorem to expand $(1 + 0.02)^5$ to the x^2 -style second-order term, and hence estimate 1.02^5 to four decimal places.

$$1 + 5(0.02) + 10(0.02)^2 + \dots = 1 + 0.1 + 0.004 + \dots \approx 1.1041 \text{ (true value } 1.10408\dots).$$

More expansions and terms

- 7 Expand $(2x - y)^3$ fully.

$$8x^3 - 12x^2y + 6xy^2 - y^3.$$

- 8 Find the middle term of the expansion of $(x + 3)^6$ (the term with x^3).

Middle term is the 4th term: $\binom{6}{3}x^3(3)^3 = 20 \cdot 27x^3 = 540x^3$.

- 9 Find the constant term in the expansion of $(x^2 + \frac{1}{x})^6$.

General term: $\binom{6}{k}(x^2)^{6-k}(\frac{1}{x})^k = \binom{6}{k}x^{12-3k}$. Constant term needs $12 - 3k = 0$, so $k = 4$: $\binom{6}{4} = 15$.

Sums of binomial coefficients

- 10 Evaluate $\binom{5}{0} + \binom{5}{1} + \binom{5}{2} + \binom{5}{3} + \binom{5}{4} + \binom{5}{5}$ without computing each term, using the identity $\sum_{k=0}^n \binom{n}{k} = 2^n$.
 $= 2^5 = 32$.
- 11 Use the fact that $\binom{n}{k} = \binom{n}{n-k}$ (symmetry of Pascal's triangle) to explain why $\binom{9}{2} = \binom{9}{7}$ without computing either value directly.
 Choosing 2 items to include from 9 is equivalent to choosing 7 items to exclude, so the two counts must be equal — this is the symmetry property $\binom{n}{k} = \binom{n}{n-k}$ with $n = 9$.

Further expansions

- 12 Find the coefficient of x^4y^3 in the expansion of $(x + 2y)^7$.
 Term: $\binom{7}{3}x^4(2y)^3 = 35 \cdot 8x^4y^3 = 280x^4y^3$. Coefficient: 280.
- 13 Find the first three terms in the expansion of $(1 - 2x)^6$ in ascending powers of x .
 $\binom{6}{0}1^6 + \binom{6}{1}1^5(-2x) + \binom{6}{2}1^4(-2x)^2 + \dots = 1 - 12x + 60x^2 - \dots$.
- 14 Use the binomial theorem with $(1 - 0.01)^4$ to approximate 0.99^4 to four decimal places, using the first three terms.
 $1 + 4(-0.01) + 6(-0.01)^2 + \dots = 1 - 0.04 + 0.0006 = 0.9606$ (true value 0.96059601).

Deriving Pascal's triangle recursively

- 15 State Pascal's Rule, $\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$, and use it to compute $\binom{6}{3}$ from row 5 of Pascal's triangle (1, 5, 10, 10, 5, 1).
 $\binom{6}{3} = \binom{5}{2} + \binom{5}{3} = 10 + 10 = 20$.

The general term formula

- 16 State the general term (the $(k + 1)$ th term) formula for the expansion of $(a + b)^n$, and use it to find the 5th term of $(x - 2)^8$ directly (without expanding the whole thing).
 General term: $T_{k+1} = \binom{n}{k}a^{n-k}b^k$. For the 5th term, $k = 4$: $T_5 = \binom{8}{4}x^4(-2)^4 = 70 \cdot x^4 \cdot 16 = 1120x^4$.

Combinatorial interpretation of binomial coefficients

- 17 Explain the combinatorial meaning of $\binom{n}{k}$, and use it to explain why $\binom{n}{0} = \binom{n}{n} = 1$ for any n .
 $\binom{n}{k}$ counts the number of ways to choose k items from a set of n (order doesn't matter). $\binom{n}{0} = 1$ because there is exactly one way to choose nothing (the empty selection), and $\binom{n}{n} = 1$ because there is exactly one way to choose everything (select the whole set).

Applying the binomial theorem to probability-style counting

- 18** A fair coin is flipped 5 times. Using $\binom{5}{k}$, find the number of outcomes with exactly 3 heads, and state the total number of possible outcomes (using 2^5).
Outcomes with exactly 3 heads: $\binom{5}{3} = 10$. Total possible outcomes: $2^5 = 32$.
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A closing challenge problem

- 19** Find the sum of all the coefficients in the expansion of $(3x - 2)^4$, using the fact that setting $x = 1$ gives the sum of all coefficients.
Substituting $x = 1$: $(3(1) - 2)^4 = (1)^4 = 1$. The sum of all coefficients in the expansion is 1.
- 20** Find the coefficient of x^5 in the expansion of $(x^2 + 3)(x - 1)^6$, by finding the relevant terms from $(x - 1)^6$ first.
From $(x - 1)^6$, the terms of degree 3 and 5 are needed (since multiplying by x^2 or by the constant 3 must produce x^5). Term with x^3 : $\binom{6}{3}x^3(-1)^3 = -20x^3$; multiplying by x^2 gives $-20x^5$. Term with x^5 : $\binom{6}{1}x^5(-1)^1 = -6x^5$; multiplying by 3 gives $-18x^5$. Total coefficient: $-20 + (-18) = -38$.
- 21** Use the binomial theorem to expand $(x + h)^3 - x^3$ fully, then divide by h and simplify — this expression is exactly the difference quotient used to find the derivative of $f(x) = x^3$ in calculus.
 $(x + h)^3 = x^3 + 3x^2h + 3xh^2 + h^3$ (binomial expansion). So $(x + h)^3 - x^3 = 3x^2h + 3xh^2 + h^3$.
Dividing by h : $\frac{3x^2h + 3xh^2 + h^3}{h} = 3x^2 + 3xh + h^2$ — as $h \rightarrow 0$, this approaches $3x^2$, the derivative of x^3 .
- 22** Evaluate $\binom{7}{2}$ and $\binom{7}{5}$, and confirm they are equal using the symmetry property.
 $\binom{7}{2} = 21$ and $\binom{7}{5} = 21$. Equal, confirming $\binom{n}{k} = \binom{n}{n-k}$ with $n = 7$, $k = 2$.